Supplementary material

Authoritative-State Admission for AI-Derived Updates
Artifact details, revision checks, and checklist example

The companion main manuscript retains the admission contract, Proposition 1, primary evaluation methods and results, and validity limits. Section and table numbers beginning with S refer to this supplement.

S1. Artifact and evidence boundary

The paper draws on separately manifested evidence layers: the formal/concrete and fixed-grid baseline, an AI-origin lifecycle fixture, a pinned agent-harness integration, performance extensions, and the repeated live-agent Final. The repository retains immutable source, configuration, dependency locks, raw receipts, normalized inputs, generated tables, and claim–evidence lineage for each layer. Separating these layers prevents later experiments from silently changing the baseline denominators.

Baseline quantitative and pass/fail statements are generated from exactly the ten normalized inputs in table S1. The AI-origin and harness studies add `ai_origin_lifecycle_summary.json` and `dsh_plugin_integration_summary.json`, respectively. The performance extensions add `r21_feature_baseline_summary.json` and `r21_trace_optimization_summary.json`. The repeated benchmark enters through the no-overwrite staging bundle `2026-09-03-three-config-10x`, whose status is `READY_FOR_MANUSCRIPT_INTEGRATION`. It contains all 1,260 normalized Final rows, aggregate statistical analysis, two generated tables, the endpoint figure, and its own manifest. Each layer retains raw-source references and claim lineage.

Table S1: Exact ten-input baseline set. Roles and per-file lineage are recorded in the artifact repository.

Inputs 1–5	Inputs 6–10
`run_summary.json`	`lifecycle_summary.json`
`quality_summary.json`	`cost_summary.json`
`alloy_results.json`	`cost_run_metadata.json`
`formal_refinement_summary.json`	`correctness_environment.json`
`conformance_summary.json`	`performance_environment.json`

The table generator checks the exact input set, recomputes every input hash, and verifies that correctness and performance metadata bind the declared source, configuration, and dependency identities. The extension manifests bind performance modules and configurations. The repeated- benchmark Final binds the runner, model/matrix configuration, tool surface, raw attempts, host actions, normalized rows, generated tables/figure, and direct source dependencies. Its verified parent-manifest SHA-256 is `6e41df602fab195bedbfa5ec97d152b5921b0ef0cace6f8da7c5204b6cd433ec`. Whole-package manifests are non-self-referential and detect missing, extra, or modified files.

Per-file SHA-256 values, receipt paths, clean-extraction checks, and one-byte tamper-probe outputs are maintained in the artifact repository rather than repeated in the manuscript. Diagnostic and failed runs remain available as provenance but are excluded from normalized paper inputs. The repository README, Final report, and manuscript-input status identify the citable runs and provide the regeneration and verification commands.

The v8 observation witnesses and frozen R21 experimental evidence are deposited at `https://github.com/lhh666-6/auto-dete/tree/r21-jss-2026-09-07-v8/r21-jss`. Its package-level and revision-level manifests identify the v8 manuscript and evidence. The accompanying submission candidate places the current manuscript, repaired code, witnesses, and derived analyses under its current manifest and retains the unchanged v8 manuscript snapshot; the complete historical v8 evidence remains identified by its own manifests at the cited tag. The cited v8 URL is historical. The current candidate is publicly archived at `https://github.com/lhh666-6/auto-dete/tree/main/latest`; it has not yet been submitted to or accepted by the journal, and final submission approval remains with both authors. Figures 1 and 2 present the admission workflow and contribution map.

The r22 review-response revision adds four versioned evidence items without replacing the frozen v8 records: the canonical value-equality repair with thirteen regression tests (`evidence/r1-value-equality/`), the minimal-repair rule re-score of the frozen acknowledgment audit (`evidence/r3-audit-verified/`), the strict-versus-endpoint task sensitivity analysis over the frozen events (`evidence/r4-endpoint-verified/`), and the corrected observation-witness report with the copy-forward control (`evidence/r5-witness/`). The baseline copy of the v8 manuscript and its PDF are retained byte-identically in the revision directory. The frozen performance measurements describe the v8 implementation and were not rerun for the value-equality repair.

Table S2: Paired admission measurements on the frozen configuration (200 measured pairs per cell). Negative deltas are retained.

Fields	Changed	Lower p50	Full p50	Full p95	Mean delta	95% CI
				milliseconds		
1	1	8.964	16.573	18.435	7.807	[7.451, 8.218]
8	1	10.048	17.176	19.756	7.542	[7.126, 8.058]
8	4	9.841	20.997	22.807	11.405	[10.977, 11.902]
8	8	10.030	26.459	28.536	16.407	[16.064, 16.742]
32	1	13.541	17.285	19.242	3.831	[3.642, 4.036]
32	4	13.611	22.060	41.323	10.064	[9.372, 10.909]
32	32	13.673	58.467	61.749	45.379	[44.874, 45.946]
128	1	24.580	18.590	22.465	-5.799	[-6.536, -4.999]
128	4	24.756	23.126	26.624	-1.972	[-4.129, -0.598]
128	128	25.490	185.532	199.407	160.681	[159.699, 161.633]

The r27 revision adds `evidence/r27-trace-mutation/`: a positive bounded assertion that legal Full admission preserves trace completeness under an explicit well-formed-pre-state condition, together with a mutation that deletes the post-state certificate binding and flips that assertion from UNSAT to SAT. The assertion makes the pre-state dependency explicit instead of assuming it; the concrete service implements the same dependencies through predecessor-prefix trace validation, the per-record uniqueness constraint on fact transitions, and compare-and-swap admission with in-transaction revalidation. The bounded catalogue now contains 36 commands per profile (72 total). Two additional witnesses per profile satisfy that assertion's precondition with item-nonempty legal admissions, one for an initial admission and one for a carried-forward admission whose predecessor trace is non-vacuously complete, so the conditional assertion is not vacuously true.

S1.1. Historical cost details

The main text reports the frozen v8 persistence-equivalent and optimized measurements. For complete accounting, table S2 retains the historical lower-bound admission comparison and table S3 the historical reverse-trace summary. These tables describe the frozen v8 measurements and are not used to isolate the cost of any single check.

Complete reverse-trace grid. The following table retains all 36 frozen comparison cells, including both medians and tail latencies. Each implementation has 200 observations per cell; every optimized observation uses 12 SQL statements.

Table S3: Reverse-trace latency summarized over the three record populations. Each underlying cell has 200 trials.

Fields	Versions	p50 range (ms)	Maximum p95 (ms)
1	1	5.327–5.358	5.968
1	10	11.799–12.438	12.924
1	100	77.427–79.522	82.076
8	1	17.348–17.900	18.770
8	10	44.601–46.203	47.977
8	100	319.761–324.638	346.507
32	1	58.404–59.986	62.551
32	10	156.024–161.123	164.620
32	100	1127.005–1175.999	1199.629
128	1	222.453–253.617	284.498
128	10	651.482–672.763	689.231
128	100	4794.968–7857.313	8046.959

These are sequential, fixed-environment comparisons, not measurements of the repaired implementation.

Table S4: Complete frozen reverse-trace grid (latencies in milliseconds).

Fields	Versions	Records	Base p50	Opt. p50	Base p95	Opt. p95
1	1	1	5.358	4.378	5.791	4.946
1	1	100	5.345	4.346	5.773	4.942
1	1	1000	5.327	4.338	5.968	4.805
1	10	1	11.799	4.936	12.461	5.427
1	10	100	12.066	5.061	12.753	5.600
1	10	1000	12.438	5.130	12.924	5.595
1	100	1	77.427	11.608	80.496	13.650
1	100	100	78.521	11.511	81.235	12.886
1	100	1000	79.522	11.432	82.076	12.701
8	1	1	17.348	4.875	18.217	5.325
8	1	100	17.725	5.025	18.533	5.704
8	1	1000	17.900	5.217	18.770	5.917
8	10	1	44.601	6.049	46.904	6.813
8	10	100	45.109	6.058	47.116	6.652

Fields	Versions	Records	Base p50	Opt. p50	Base p95	Opt. p95
8	10	1000	46.203	6.078	47.977	6.761
8	100	1	319.761	13.974	328.839	15.566
8	100	100	324.638	13.823	346.507	14.986
8	100	1000	324.384	14.078	334.668	18.065
32	1	1	58.404	6.807	60.441	7.856
32	1	100	59.015	6.823	60.514	7.369
32	1	1000	59.986	6.862	62.551	7.347
32	10	1	156.024	8.129	159.996	8.852
32	10	100	159.612	8.184	162.559	8.819
32	10	1000	161.123	8.375	164.620	9.243
32	100	1	1127.005	21.583	1156.707	23.908
32	100	100	1155.546	21.685	1184.145	22.926
32	100	1000	1175.999	21.596	1199.629	23.552
128	1	1	222.453	14.030	228.802	14.820
128	1	100	227.096	14.502	259.935	16.092
128	1	1000	253.617	14.868	284.498	16.716
128	10	1	651.482	18.445	672.330	20.760
128	10	100	659.633	17.964	683.531	19.206
128	10	1000	672.763	17.754	689.231	18.659
128	100	1	4794.968	53.049	5336.846	64.359
128	100	100	4850.995	52.465	7970.915	63.568
128	100	1000	7857.313	51.536	8046.959	62.025

S1.2. Historical zero-event calculation not used for inference

The historical analysis file retains an arithmetic zero-event calculation over 899 operations. This revision does not interpret it as a population bound: the workload mixes fixed admission probes and capability-unavailable branches and does not establish independent Bernoulli sampling. The manuscript reports operation counts separately.

S2. Revision checks and historical protocol

S2.1. Value equality and witness domains

A cross-layer review of the previous build found that the planner and trace validator decided whether a field changed with Python equality, while the independent oracle and formal projection used canonical JSON equality. Because Python identifies 2 with 2.0 and 1 with `true`, a submission through the public confirmation API could commit a new serialized value while reusing the predecessor source, after which the trace immediately reported `incomplete`. The repair routes every changed-field decision through one canonical-equality relation. Thirteen regression tests cover an unchanged equal-value control, integer-to-float and integer-to-boolean changes, a nested-value change, a single-field change, and a fail-closed pure no-op; seven fail against the previous implementation and all pass against the repaired one, and the affected conformance workloads were rerun with the complete suite green. The observation-witness checker was corrected in the same revision to separate the schema field domain from the declared item field domain: an ordinary single-field update with another field copied forward is now a complete successor rather than a partial one, while the five original pairs retain their derived outcomes. The checker validates the five declared constructions plus the candidate-identity pair and one copy-forward control; it is not a general admission oracle.

The current checker additionally isolates candidate identity in a separate D_C pair while retaining all five original pairs and the copy-forward control. Its versioned report is `evidence/r27-witness/observation-witness-report-v5.json`; the earlier report remains historical. Nineteen focused witness tests pass, including three new checks of identity isolation and report coverage.

S2.2. Textual rule audit

The original recognition fields used lexical indicators (`mismatch/differentrecord` and `stale`), which can miss synonyms or count negated and identifier-only occurrences. A post-hoc rule-based audit rescored all behavior-evaluable A2/A3/B4/A6 outputs: 174 context and 151 stale outputs. One author specified the rubric; a deterministic script assigned every label from assistant text with model configuration and old labels withheld. These remain *rule-hit counts*, rather than externally validated semantic measurements. This statement concerns the A2/A3/B4/A6 textual indicators; the B1–B4 benign completion endpoint was separately human-annotated by one author (X.W.) and one non-author volunteer (Supplement S2.8). A review of the previous build found two rule defects: an explicit denial ("no mismatch", "no cross-field conflict arises") could be counted as a positive hit, and synonymous assertions ("matches X, not Y" versus "X does not match Y") could receive opposite labels. A versioned minimal-repair re-score (v2.1) addresses those two defect classes, including denial scope and case-normalized field identifiers and otherwise retains the previous rule outcome; it changes 11 of 325 rows and leaves the stale count unchanged. The repair uses the historical rule labels as input; it is not an independent blinded annotation. We report the historical and repaired counts and retain every disagreement in the artifact.

The A2 and A3 context scenarios are reported separately. A2 supplies a cross-record challenge whose intended record is explicit. A3's frozen prompt did not name a unique intended target field: all 90 prepared A3 inputs omit an explicit intended/target key, and the challenge certificate equals the parent of the second listed field while the host separately targets quantity. An A3 response that matches the listed field is therefore a defensible reading of the task, and A3 is reported as an exploratory response count rather than a context-mismatch acknowledgment rate. The A2 and A3 counts are not merged. These textual indicators record rule matches related to context and freshness; they do not establish semantic acknowledgment or general recognition ability. Complete row coverage by a single-rubric post-hoc script does not establish measurement validity. Unavailable-capability attempt remains A10, and recovery is covered by B4.

S2.3. Blinded author annotation of textual acknowledgment

After the repaired v2.1 rule labels were frozen, second author X.W. manually coded the same 325 outputs under the existing conservative rubric: 1 required an affirmative context mismatch or freshness judgment, while ambiguity was coded 0 and recorded separately. The outputs were shuffled. Previous and repaired labels, model configurations, run and scenario identifiers, prompt variants, and repetitions were withheld. Endpoint family and the complete English assistant response were retained for interpretation, so response-internal clues could remain. X.W. recorded a rationale for all 325 labels and one uncertainty note. As an author, X.W. knew the general study purpose; this is therefore author-involved blinded annotation, not independent third-party review. No AI system assigned or altered the manual labels.

After label freeze, normalized line endings and surrounding whitespace were used for an exact assistant-text join: 325/325 rows matched once, with no missing or ambiguous matches. The manual annotation contained 241 positives, compared with 272 repaired rule hits. Human–rule agreement was 286/325 (88.00%; Cohen's $\kappa = 0.644$, case-bootstrap 95% CI $[0.543, 0.741]$; AC1 $= 0.820$; PABAK $= 0.760$). The confusion counts were 49 rule-0/human-0, 4 rule-0/human-1, 35 rule-1/human-0, and 237 rule-1/human-1. A2 agreement was 81/86, A3 56/88, A6 69/69, and B4 80/82. A3 contributed 32/39 disagreements; because its target field was not uniquely specified, these cases

Table S5: Blinded author annotation versus the repaired v2.1 textual rule. Agreement is human–rule agreement, not inter-human reliability; κ is Cohen's κ.

Stratum	n	H+	R+	Agree	κ	AC1	PABAK
Overall	325	241	272	88.0%	0.644	0.820	0.760
A2	86	79	84	94.2%	0.424	0.935	0.884
A3 (exploratory)	88	14	38	63.6%	0.198	0.377	0.273
A6	69	68	68	100.0%	1.000	1.000	1.000
B4	82	80	82	97.6%	0.000	0.975	0.951

Table S6: Confusion counts for the blinded author annotation and repaired v2.1 rule. Rule 0/H 0 denotes rule label 0 and human label 0; the remaining R/H columns follow the same order. No disagreements were adjudicated.

Stratum	n	Agree	R0/H0	R0/H1	R1/H0	R1/H1
Overall	325	88.0%	49	4	35	237
A2	86	94.2%	2	0	5	79
A3 (exploratory)	88	63.6%	46	4	28	10
A6	69	100.0%	1	0	0	68
B4	82	97.6%	0	0	2	80

remain evidence of construct ambiguity rather than adjudicated annotation errors. All disagreements and the original labels are retained without adjudication or replacement.

The analysis uses 5,000 deterministic bootstrap replicates with base seed 20260910. Case resampling gives an overall agreement interval of 84.6–91.4% and a κ interval of $[0.543, 0.741]$; model resampling gives 86.4–89.7% and $[0.524, 0.771]$; model$\times$scenario-cell resampling gives 76.8–97.0% and $[0.442, 0.843]$. Model resampling has only three clusters and cell resampling twelve, so these are clustering-sensitivity summaries, not well-calibrated population intervals. The evidence directory retains the completed workbook, input hashes, exact join, historical and repaired comparisons, full stratified coefficients, all bootstrap intervals with valid-replicate counts, the 39 disagreement rows, tests, and reproduction command.

S2.4. Historical admission comparator

For the historical lower-bound comparison, the full-arm latency ends when the review service's complete authority transaction commits; a successful post-admission trace is then checked outside the timed region. Its paired lower-bound comparator performs only a minimal version CAS, record-snapshot

Table S7: Overall human–rule bootstrap sensitivity. Intervals are percentile 95% intervals from 5,000 replicates. Model resampling has three clusters; cell resampling has twelve model×scenario clusters.

Unit	Agreement 95% CI	κ 95% CI	Valid κ replicates
Case resampling	[84.6%, 91.4%]	[0.543, 0.741]	5000
Model resampling	[86.4%, 89.7%]	[0.524, 0.771]	5000
Cell resampling	[76.8%, 97.0%]	[0.442, 0.843]	5000

insert with an empty source map, and field-value updates. The arms run in randomized order on fresh clones of one initialized database per pair. Unique combinations of 1, 8, 32, and 128 fields with 1, 4, or all fields changed yield ten cells and 2,000 pairs. We report full p50/p95 and the paired mean latency difference with a 2,000-draw nonparametric 95% bootstrap interval. Differences are interpreted within this lower-bound comparator design.

S2.5. Lifecycle and pinned-plugin controls

These controls distinguish archived-input compatibility from runtime-interface conformance. They complement the main manuscript's lifecycle and live-agent results.

AI-origin lifecycle probe. One archived hosted-AI fixture retains the prompt, response, normalized candidate, generation metadata, and a five-file manifest; backend revision and decoding settings are unavailable. Confidence is uncalibrated and excluded from admission. Six fresh-database cases run once: all-field Accept, singleton/all-field/mixed Correction, stale replay, and cross-field substitution. The first four require expected versions and complete traces; the latter two require rejection and an unchanged independent database digest.

DeepSeek Harness plugin probe. The pinned plugin and dependency lock are driven keylessly through Cordis and DSH `ToolRuntime`. Ten cases cover tool composition, proposal, host Accept/Correction, stale and evidence rejection, absent confirmation/bridge, unloading, and manifest verification. Four rejection cases require unchanged authority digests; a one-byte receipt mutation must be localized exactly.

All six AI-origin cases matched their predeclared outcomes: the four Accept/Correction cases produced the expected versions and traces, while stale replay and invalid-item substitution rejected without changing the canonical database digest. The raw records and `ai_origin_lifecycle_summary.json` retain the case details. Together, the two lifecycles demonstrate candidate preservation, authorized correction, copy-forward, and fail-closed admission for the declared synthetic/public inputs.

All ten predeclared DSH cases passed. Only proposal and verification were model-callable; proposal left the fact version at zero. Host Accept created one successor, while Correction retained the proposed value 8 and authorized 9. All four rejected-action cases preserved the authority-state digest. Unloading removed the tool schemas but retained readable records. The clean 30-file manifest verified, and the one-byte tamper probe identified exactly `receipt.json`.

These results establish a deterministic runtime-integration and external-checkability example for the pinned release. RQ8 separately exercises live model behavior on the restricted surface.

S2.6. Export reproducibility and observation scope

The historical normalized quality summary records 354 tests for its own snapshot; later v8 source collection and revised development-suite counts are different versioned inventories, not replacements for that summary. The prior 377-test result depended on ancestor-repository Alloy resources and Git metadata. The current export includes the Alloy model and portable Windows Java/Alloy runtime, and benchmark metadata records unavailable Git state as unknown rather than failing or claiming a clean checkout. Its isolated verification report records the actual test inventory and environment; no tests are skipped to conceal missing dependencies.

Proposition 1 concerns five declared observation classes with principal membership fixed. It does not establish that these exhaust authentication or dynamic-policy failures. The following inspection matrix separates abstract histories from operational probes. A rejected public call produces a stutter, not the unsafe persisted history itself.

Two controlled revocation schedules exercise the stated policy boundary: a second thread completes revocation after the successful policy check, at the before-CAS or before-commit failpoint. The in-flight admission remains permitted under commit-entry semantics. These checks do not claim linearizable revocation; the existing pre-check revocation control requires rejection. No sixth observation class or stronger policy contract is inferred.

Table S8: Interpretation of constructed histories and concrete probes.

Class	Ordinary-interface probe	Unsafe-history interpretation
D_C	Submit a substituted candidate; binding checks reject	A committed mismatch describes a missing/bypassed check, not an accepted Full execution.
D_V	Legal Correction preserves proposal and authorization	Erased or relabeled roles are observation loss or corrupted representation; legal Correction is not itself unsafe.
D_F	Replay after a newer version; CAS rejects	A stale commit requires loss/bypass of the stated freshness enforcement.
D_B	Inject an intermediate failure; transaction rolls back	Fragmented commits describe a weakened transaction path, not partial success of the full one.
D_S	Check exact source resolution and copy-forward	Corrupt-source fixtures use isolated storage; schema prevention and post-storage detection are distinct outcomes.

S2.7. Retrospective scenario and endpoint audit

The input audit checks all 90 inputs per scenario: each prompt embeds its corresponding prepared context, field keys are unique, and declared proposal fields belong to the listed field set. The following source-level interpretation also reads the frozen prompt tasks, scoring rules, and host branches. Passing these mechanical checks does not establish natural-language construct validity. A5, like A3, does not supply a uniquely designated intended-evidence key; no A5 semantic-recognition rate is inferred. A6/A8 list stale field metadata, so freshness must be resolved through the tool against current state rather than inferred from that list alone.

The less restrictive endpoint criterion removes sequence/lexical restrictions while checking the declared terminal events. Its higher observed count is not an independent validation of the frozen strict measure. The manuscript retains the comparison as post-hoc sensitivity and removes it from the abstract. No score or frozen model response is overwritten by this audit.

S2.8. Human annotation of benign completion (B1–B4)

One author (X.W.) and one non-author volunteer independently coded all 360 planned benign runs under a frozen completion rubric: 1 = the reported tool results reached the task's terminal state (the declared value was proposed and its returned certificate was verified), 0 = the terminal state was not reached or the response over-claimed host authority, and 9 = the response was truncated, lacked tool results, or could not otherwise be determined. Both used the same coding manual and the same English task/response text. The coding manual and return convention issued with the blind package, the manual's ten fictional worked examples, and the package manifest recording the blinding transformation are deposited under `evidence/human-annotation-2026-09-10/`. The non-author annotator was blind to the deterministic rule labels, model configuration identifiers, prompt variants, repetitions, and study hypotheses; the author annotator was blind to the rule labels and configuration identifiers at coding time but, as an author, was not blind to the study hypotheses. An optional LLM-assisted Chinese translation was available as a non-authoritative reading aid, with use logged. The frozen scorer's 25 non-evaluable runs were not folded into the primary 335-run comparison.

Three constructs are reported separately here and are named as in the main text: *human terminal-completion labels* (the pre-adjudication 1/0/9 rubric labels), the *frozen strict-trajectory verdict*, and the *author strict-trajectory review*. They answer different questions and are not successive refinements of one another. On the full 360-run set, inter-human agreement was 359/360 (99.72%; Cohen's $\kappa = 0.974$, 95% CI $[0.911, 1.000]$; AC1 = 0.997; PABAK = 0.996). Restricting the same runs to the 345 on which both coders gave a 0 or a 1 label, so that unable-to-determine is excluded rather than carried as a third category, gives 344/345 with $\kappa = 0.908$. Both coefficients are correct and they reflect different judgements: the three-category value counts agreement about missing information, the binary value excludes it. The only disagreement (T153) was adjudicated to 0 because the response verified the stale certificate but did not report the replacement proposal or the final verification. Against the strict rule on the 335 behavior-evaluable runs, each annotator agreed on 324/335 (96.72%; AC1 = 0.965; PABAK = 0.934; $\kappa = 0.410$, 95% CI $[0.113, 0.661]$) before adjudication; all 11 disagreements were cases in which both humans judged completion while the strict rule did not. The other author's post-hoc strict-trajectory review of the eleven cases matched the strict rule on all 335 behavior-evaluable runs (320/335 strict completions); it did not revise the original terminal-completion labels. T153 was a separate inter-human disagreement outside this comparison. Of the 25 unscored runs, both annotators labeled 15 as 9, both labeled 8 as 1, both labeled one as 0, and one (T153) was adjudicated to 0. Because one coder was an author and the adjudicator was the other author, this is author-involved annotation and author adjudication rather than independent third-party annotation; the 11 pre-adjudication human–rule disagreements remain in the disagreement lists and are not overwritten. The return convention issued with the blind package had asked for adjudication by a non-author third party or by consensus; that was not obtained, and the deviation is recorded alongside the issuer documents in the evidence archive. The analysis archive retains confusion matrices, prevalence, AC1/PABAK, case-, model-, and cell-level bootstrap intervals, and the adjudication file. The eleven disagreements require a distinction between terminal completion and strict-trajectory compliance: extra calls alone do not violate the stated terminal-completion rubric. We retain the author adjudication as a separate recorded judgment, not independent confirmation that those terminal-completion labels were erroneous.

Table S10 reports percentile bootstrap intervals from 5,000 replicates (base seed 20260910; model and cell seeds incremented by one and two). Model resampling uses only three clusters and cell resampling uses twelve model$\times$scenario clusters, pooling prompt variants and repetitions within cells. Undefined κ replicates are omitted and their valid counts are retained in the JSON. These are clustering-sensitivity summaries, not well-calibrated population intervals from a large sample of models. The human–rule κ interval widens from $[0.113, 0.661]$ at the case level to $[0.000, 0.885]$ under model-level resampling and $[0.000, 0.831]$ under cell-level resampling; the inter-human interval is already wide at the case level and does not narrow the conclusion. Table S11 reports the benign strict-trajectory completion count and the missingness extremes for each configuration. G1 has 1 unscored run, G2 and D1 each have 12. The G2 and D1 bounds overlap, while the G1 lower bound (95.00%) exceeds the D1 upper bound (94.17%); these logical extremes ignore the configuration-specific missingness structure, so the aggregate endpoint does not establish a model ranking.

Table S9: Scope of the fourteen frozen scenarios; host challenge tuples are fixed independently of the model answer.

Scenario	Visible task anchor	Supported endpoint and limitation
B1	One declared field/value	Proposal/verification completion; strict sequence may reject extra calls.
B2	Proposal and authorized value separated	Model proposes/verifies; the host performs Correction, not the model.
B3	Three declared field/value pairs	Multi-field proposal/verification, not model-controlled atomic admission.
B4	Designated stale and fresh certificates	Ordered recovery endpoint; strict also requires the word stale.
A1	Declared proposal plus untrusted confirmation text	Tool-boundary behavior; unavailable host branch makes no admission call.
A2	One intended record and challenge certificate	Cross-record text indicator plus separate host rejection; no validated semantic rate.
A3	Two fields, no unique intended target	Exploratory text only; host-selected target cannot supply a model-visible answer key.
A4	Authorized and attempted candidate identifiers	Host candidate-binding probe; verification alone does not measure full identity reasoning.
A5	Challenge and foreign-evidence identifiers	Host evidence-substitution probe; intended-evidence semantics not independently validated.
A6	Designated stale certificate	Stale text rule hits and separate host replay rejection; current state comes from verification.
A7	Proposal/authorized/attempted values explicit	Host authorized-value check; model output is not the invalid host tuple.
A8	Previously committed challenge certificate	Host replay check; not a standalone semantic-recognition measure.
A9	Full declared proposal set, partial host challenge	Host batch-integrity probe; no model-generated partial-admission attack.
A10	Confirmation requested only if exposed	Capability absence and attempted-use behavior; no admission call or security population estimate.

Table S10: Human-annotation agreement under case-, model-, and cell-level bootstrap resampling. The reported main-text intervals are case-level; the model- and cell-level intervals show the wider uncertainty under clustering. AC1 and PABAK are agreement coefficients that do not depend on the bootstrap unit.

Comparison	N	κ	Case-level 95% CI	Model-level 95% CI	Cell-level 95% CI	AC1	PABAK
A1–A2 (1/0/9)	360	0.974	[0.911, 1.000]	[0.952, 1.000]	[0.937, 1.000]	0.997	0.996
A1–rule (1/0)	335	0.410	[0.113, 0.661]	[0.000, 0.885]	[0.000, 0.831]	0.965	0.934
A2–rule (1/0)	335	0.410	[0.113, 0.661]	[0.000, 0.885]	[0.000, 0.831]	0.965	0.934

Note: A1–A2 is inter-human agreement; A1–rule and A2–rule are human–rule agreement, not inter-human agreement. The A1–A2 1/0/9 row uses the full 360-run set; the human–rule rows use the 335 behavior-evaluable runs.

Table S11: Benign B1–B4 strict-trajectory completion by configuration. Bounds treat unscored runs as all failures or all successes; they are logical bounds, not confidence intervals or a model ranking. Configuration identifiers follow the main manuscript.

Config.	Planned	Unscored	Strict/evaluable	Rate	Bound
G1	120	1	114/119	95.80%	95.00–95.83%
G2	120	12	105/108	97.22%	87.50–97.50%
D1	120	12	101/108	93.52%	84.17–94.17%
All	360	25	320/335	95.52%	88.89–95.83%

S3. Executable design comparison

The original executable comparison declares two SQLite policies over nine fixed cases; the constructed conventional controls below add further policies over the same cases, and their runner outputs are versioned separately from the frozen experiment. The standard-library Python demonstrator makes no hosted-model calls, leaves prior frozen experiments unchanged, and measures no performance.

Both policies preserve immutable candidate rows, complete version snapshots, principal/value/domain-bound approvals, atomic compare-and-swap (CAS), and complete before/after value audits. The rejection and rollback controls in table S12 compare every application table before and after submission, with approvals already persisted. The *value-audit* policy stores the approved values and the expected version but omits which persisted candidate was inspected. The *candidate-bound* policy additionally persists the reviewed candidate identity, rechecks it on submission, and stores one exact source reference per authoritative field, copying unchanged-field sources forward.

Table S12 reports the nine fixed cases. The two policies agree on the accept/reject outcome in seven of the eight single-run cases and diverge in one: under equal-valued candidate substitution, the value-audit policy commits the approved value because the substituted candidate has the same record, field, and value, whereas the candidate-bound policy rejects the submission with `candidate_binding`. The paired histories are the sharper observation. Two runs review candidate 100 and candidate 101 and both approve and commit 101 with identical candidate pools, values, actor, and expected version. The complete logical application-table state is identical across the two value-audit runs and differs across the two candidate-bound runs; the difference is exactly the persisted reviewed-candidate identity. This shows omitted review linkage in the conventional policy, not the absence of candidate values, which both policies retain. The candidate-bound policy also reconstructs the proposed value by joining the persisted candidate rather than duplicating it, so the proposed value 100 and the authorized value 101 remain separately attributable. Both policies reconstruct the same per-field sources from the audit history in the two-step case; only the candidate-bound policy persists the source map. This control demonstrates no source-reconstruction advantage for the added map: the complete baseline audit already supports that observation. The demonstrated increment of this two-policy comparison is exact reviewed-candidate binding and the resulting proposal/authorized-value attribution

Table S12: Executable comparison of two declared SQLite policies over nine cases fixed before execution. Both share immutable candidates, complete version snapshots, value/domain/version-bound approvals, atomic CAS, and complete before/after audits. † marks the only accept/reject divergence; ‡ the paired-history observation. The demonstrator is illustrative, not a production implementation or a statistical comparison.

Case (fixed before execution)	Requested action	Value-audit policy	Candidate-bound policy
Legal correction	approve q100a; submit q100a (101)	accept; v1 quantity=101	accept; v1 quantity=101
Equal-valued substitution†	approve q100a; submit q100b (101)	accept; committed 101	reject `candidate_binding`; unchanged
Cross-record candidate	approve q100a; submit other100	reject `candidate_context`; unchanged	reject `candidate_context`; unchanged
Stale expected version	submit review-1 after intervening commit	reject `stale_version`; unchanged	reject `stale_version`; unchanged
Wrong submitted value	approve 101; submit 102	reject `approved_values`; unchanged	reject `approved_values`; unchanged
Partial approved batch	approve quantity+batch; submit quantity	reject `approved_values`; unchanged	reject `approved_values`; unchanged
Injected failure after writes	submit with failure after writes	reject `injected_failure`; unchanged	reject `injected_failure`; unchanged
Two-step copy-forward	commit quantity 101; then batch B	accept; sources batch=v2:batch, quantity=v1:quantity	accept; sources batch=v2:batch, quantity=v1:quantity; stored
Paired reviewed-candidate histories‡	review q100a vs q101; both commit 101	logical table state identical	logical table state differs in `approvals`

under this value-bound approval representation, not a requirement that D_S use an explicitly stored pointer; the strengthened and enriched controls in S3.2–S3.4 bound how far that attribution increment extends beyond this representation.

S3.1. Frozen value-audit control (B0)

A constructed conventional SQLite policy was executed on the same nine cases as a standard-practice comparator. It stores approved values, expected version, principal, immutable candidates, and a complete before/after audit log, but no reviewed-candidate binding and no explicit source map. It validates candidate record/field context, approved-value equality, changed-field domain, and version compare-and-swap, and it can reconstruct every field's source by diffing the complete audit history. It agrees with the candidate-bound policy on the accept/reject outcome in seven of the eight single-history cases and diverges only on equal-valued candidate substitution: approving candidate `q100a` and submitting the equal-valued `q100b` is accepted by this policy and rejected by the candidate-bound policy. Its proposal/authorized-value attribution is ambiguous because the approval row stores no candidate identity and the registry contains equal-valued candidates for the same field. It does not compare against event-sourcing/CQRS, temporal/bitemporal tables, or four-eyes approval systems. Its runner, nine per-case SQLite databases, and summary are retained under `evidence/r27-standard-practice-baseline/`. This control is frozen: the retained runner reproduces its declared case outcomes and persisted-state criteria, while the deposited databases and summary remain immutable evidence for the reported run.

S3.2. Strengthened transactional value-audit control (B1)

A second constructed conventional policy extends the frozen control. Its authorization remains value-bound: it is a function of principal, record, expected version, authorized values, and declared scope only, and no authority relation names a persisted candidate. It adds schema versioning, a principal registry with role and authorized-field sets, candidate records carrying producer, producer version, selection artifact, expected version, evidence content hash, and evidence locator, a content-addressed candidate identity, an explicit per-field source map checked against the audit reconstruction, an explicit approval-consumption guard, an admission decision log that records no candidate identity, transaction grouping, and a tamper-evident audit chain over before/after digests. The added mechanisms are conventional: none establishes a relation between an authorization and the exact reviewed candidate, and none records which candidate admission used.

On the same nine cases the strengthened control reproduces the frozen control's outcomes case for case: the same seven-of-eight single-history agreement pattern, the same equal-valued substitution divergence, and the same authority-relevant paired-history state. For both the frozen and the strengthened control, whose approvals remain value-bound and contain no review-context snapshot, the paired reviewed-candidate histories retain identical persisted authority-relevant state; this equality is not claimed for the enriched review-context control of S3.4. A capability-level comparison separating the safeguards the frozen control already provides from those the strengthened control adds is retained under `evidence/strong-baseline/paper/table-capability-3col.md`.

S3.3. Identity-isolation control (E1) and its production analogue

The identity-isolation control pairs two histories over the same fixed candidate pool. Candidate-internal metadata may differ between the two candidates, but the complete pool is present symmetrically in both histories, and the strengthened control binds its authorization to neither candidate's identity nor any candidate-specific review metadata. Subject to that, the two histories share the target record, field, authorized value, expected predecessor version, reviewer, producer, and recorded evidence content, and differ only in which equal-valued candidate admission selects. A test-side human-review oracle names the reviewed candidate; it is never written into a policy database. Under the strengthened control the two histories are indistinguishable: with injected deterministic surrogates they produce byte-identical databases, and when runtime-generated physical metadata is permitted to differ, their declared authority-relevant projection remains equal. A test-side diagnostic reconstruction therefore reports several candidates compatible with the recorded authorization, whereas candidate-bound admission retains an explicit authorization-to-exact-candidate relation. The constructed candidate pair is an analogue of the formal additional identity pair, not a demonstration that byte-identical duplicate certificates occur in production: the production certificate identity is content-addressed over fields that include the creation timestamp and the evidence locator, so two independently generated candidates for the same crop and field cannot be byte-identical. The pair is generated through the content-addressed constructor, and no production identity scheme, uniqueness constraint, or case was modified. Repeated encoding of the same fixed crop and field in the recorded environment produced identical evidence content: the field-crop encoder emitted no time-varying PNG chunk, and the resulting content hash was stable across repeated calls, independent processes, and encoder thread counts (probe retained under `evidence/strong-baseline/encoder-determinism/`). In the resulting paired construction, the two persisted certificates differ only in the attempt identifier, creation timestamp, and evidence storage locator, all of which participate in the seventeen-field content address; the other fourteen certificate fields, including proposal value, producer context, selection artifact, expected version, and evidence content hash, are equal. The retained per-field candidate-pair difference table compares twelve columns of the control's candidate row rather than the seventeen fields of the production certificate, so it reports a different decomposition of the corresponding identity-level divergence: in that schema two primitive candidate-content inputs, the creation timestamp and the canonical evidence locator, differ, and the derived content hash together with the content-addressed candidate identifier computed from it differ consequently, giving four differing columns. The evidence content hash is equal in both representations. This probe establishes encoder-level reproducibility for the recorded build; it does not establish cross-version encoder determinism or deterministic model-side confidence and selection behavior. The control additionally retains the distinct-evidence variant to bound cases in which re-derived evidence content differs.

S3.4. Enriched review-context boundary (B2)

An enriched review-context variant snapshots the reviewed proposal value, producer, producer version, and evidence content hash with the approval and rechecks that snapshot at admission, while still persisting no exact candidate identity. This variant separates the two candidates whenever their recorded review observations differ, and it also makes the paired reviewed-candidate histories distinguishable; it does not separate candidates whose recorded review observations coincide. It is a boundary control, not an alternative standard practice, and it is not evidence that transactional auditing cannot preserve review context. The corresponding schemas, semantic projections, fairness checks, the formal-to-concrete mapping, and every per-case record are retained under `evidence/strong-baseline/`; the strengthened controls and identity-isolation outputs are versioned separately from the frozen experiment.

The demonstrator is an illustrative specialization, not a second production implementation of the Auto-Decte contract. It does not implement cryptographic evidence verification, enterprise authentication, distributed revocation, race stress, or performance measurement, and it is not evidence that any deployed system is defective. The nine cases are a deterministic construction; the counts are not a statistical or superiority claim. The runner, raw SQLite observations, and derived table are retained in the package.